

dMAT • CORE MODULE

Core Module Diagnostic

Mixed Timed Test

20 questions • 25 minutes • no notes • guess if unsure

Figure Sequences • Mathematical Equations • Latin Squares

These are dMAT Germany Core Module Sample Questions created by Jamboree.



How this diagnostic works

A single timed mock that samples all three Core Module task types back to back — a taste of real exam conditions.

- One clock for the whole test: 25 minutes for all 20 questions — about 75 seconds each. Keep moving.
- Three subtests run in order: Figure Sequences (7), Mathematical Equations (7), Latin Squares (6). A short instruction slide introduces each.
- Questions are unlabelled by difficulty on purpose — just like the real exam. The mix is 7 Low, 7 Medium, 6 High overall.
- No notes, no calculator. If you are unsure, make your best guess and move on — there is no penalty for guessing.
- Mark your answers on paper. When time is up, check the answer key, then study the full worked solutions at the end.

Subtest 1 — Figure Sequences

You see a series of pictures (matrices). A figure changes its position, colour and/or orientation from one matrix to the next by fixed rules. Work out the pattern and decide what the next two matrices (Matrix 5 and Matrix 6) look like.

How it works

- Figures can change colour, rotate about their own axis, and move within the grid (horizontal, vertical or diagonal).
- A rule can also grow by $x + 1$: e.g. move 1 field, then 2, then 3, and so on from matrix to matrix.
- Figures never disappear, overlap or leave the grid — at a border they move along it or bounce back.
- For each question, choose one option (1, 2 or 3) for Matrix 5 and one for Matrix 6. Both must be right to score the point.

In the real exam: 20 matrix series in 25 minutes. Be quick and accurate; guess if unsure. No notes.

Figure Sequences

Study Matrices 1-4, then choose the correct option for Matrix 5 and for Matrix 6.

Matrix 1

Yellow hexagon		
		Black triangle pointing left

Matrix 2

	Yellow hexagon	
	Black triangle pointing left	

Matrix 3

		Yellow hexagon
Black triangle pointing left		

Matrix 4

Black triangle pointing left		
		Yellow hexagon

Matrix 5

	?	

↓

Options for Matrix 5

	Black triangle pointing left	
	1	
		Yellow hexagon

	Black triangle pointing left	
		Yellow hexagon

Black triangle pointing left		
	2	
		Yellow hexagon

	Pink triangle pointing right	
		Yellow hexagon

Pink triangle pointing left		
	3	
		Yellow hexagon

	Black triangle pointing left	
		Yellow hexagon

Matrix 6

	?	

↓

Options for Matrix 6

	Black triangle pointing left	
		Yellow hexagon

	Pink triangle pointing right	
		Yellow hexagon

	Black triangle pointing left	
		Yellow hexagon

Figure Sequences

Study Matrices 1-4, then choose the correct option for Matrix 5 and for Matrix 6.

Matrix 1

		⬡
⬢		

Matrix 2

⬢		⬡

Matrix 3

⬢		
		⬡

Matrix 4

	⬢	
	⬡	

Matrix 5

	?	

↓

Options for Matrix 5

1

		⬢
	1	
⬡		

2

		⬢
	2	
⬡		

3

	⬢	
	3	
⬡		

Matrix 6

	?	

↓

Options for Matrix 6

		⬢
⬡		

⬡		⬢

⬡		⬢

Figure Sequences

Study Matrices 1-4, then choose the correct option for Matrix 5 and for Matrix 6.

Matrix 1

Yellow hexagon		
		Black triangle pointing left
Black L-shape		

Matrix 2

	Yellow hexagon	
Black L-shape		
		Pink triangle pointing right

Matrix 3

Black L-shape		Yellow hexagon
	Black triangle pointing left	

Matrix 4

	Black L-shape	
		Yellow hexagon
Pink triangle pointing right		

Matrix 5

	?	

↓

Options for Matrix 5

1

		Black L-shape
Black triangle pointing left		
		Yellow hexagon

2

Black triangle pointing left		Black L-shape
		Yellow hexagon

3

		Black L-shape
Black triangle pointing left		
	Yellow hexagon	

Matrix 6

	?	

↓

Options for Matrix 6

Pink triangle pointing right		
		Black L-shape
	Yellow hexagon	

Pink triangle pointing right		
		Black L-shape
	Yellow hexagon	

Pink triangle pointing right		
		Black L-shape
	Yellow hexagon	

Figure Sequences

Study Matrices 1-4, then choose the correct option for Matrix 5 and for Matrix 6.

Matrix 1

	Hexagon	
L		
		Square

Matrix 2

L		Hexagon
	Square	

Matrix 3

	L	
		Hexagon
Square		

Matrix 4

		L
Square		
		Hexagon

Matrix 5

	?	

↓

Options for Matrix 5

1

	Square	
		L
	Hexagon	

2

Square		
		L
	Hexagon	

3

Square		
		L
	Hexagon	

Matrix 6

	?	

↓

Options for Matrix 6

	Square	
Hexagon		L

	Square	
		L
Hexagon		

		Square
Hexagon		L

Figure Sequences

Study Matrices 1-4, then choose the correct option for Matrix 5 and for Matrix 6.

Matrix 1

Yellow hexagon		Pink triangle pointing left
		Green diamond

Matrix 2

		Yellow hexagon
Green diamond		Pink triangle pointing up

Matrix 3

Green diamond		
Pink triangle pointing right		Yellow hexagon

Matrix 4

Pink triangle pointing down		Green diamond
Yellow hexagon		

Matrix 5

	?	

↓

Options for Matrix 5

1

	Yellow hexagon	Pink triangle pointing left
		Green diamond

2

Yellow hexagon		
		Pink triangle pointing right
		Green diamond

3

Yellow hexagon		Pink triangle pointing left
		Green diamond

Matrix 6

	?	

↓

Options for Matrix 6

		Yellow hexagon
Green diamond		Pink triangle pointing up

		Yellow hexagon
Green diamond		Black triangle pointing up

		Yellow hexagon
Green diamond	Pink triangle pointing up	

Figure Sequences

Study Matrices 1-4, then choose the correct option for Matrix 5 and for Matrix 6.

Matrix 1

Yellow hexagon		Black triangle pointing left
Blue square		Black L-shape

Matrix 2

	Yellow hexagon	
Blue square		Pink triangle pointing right
	Black L-shape	

Matrix 3

Blue square		Yellow hexagon
Black L-shape		Black triangle pointing left

Matrix 4

	Blue square	
Black L-shape		Yellow hexagon
	Pink triangle pointing left	

Matrix 5

	?	

↓

Options for Matrix 5

1

		Blue square
Black L-shape		
Black triangle pointing left		Yellow hexagon

2

Black L-shape		Blue square
Black triangle pointing left		Yellow hexagon

3

Black L-shape		Blue square
Black triangle pointing down		Yellow hexagon

Matrix 6

	?	

↓

Options for Matrix 6

	Black L-shape	
Pink triangle pointing left		
	Yellow hexagon	Blue square

	Black L-shape	
Pink triangle pointing left		Blue square
	Yellow hexagon	

	Black L-shape	
Pink triangle pointing down		Blue square
	Yellow hexagon	

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Figure Sequences

Study Matrices 1-4, then choose the correct option for Matrix 5 and for Matrix 6.

Matrix 1

Yellow hexagon		Black left-pointing triangle
Orange circle		Black L-shape

Matrix 2

Blue circle		Yellow hexagon
Black L-shape		Black upward-pointing triangle

Matrix 3

Black L-shape		Green circle
Black right-pointing triangle		Yellow hexagon

Matrix 4

Black downward-pointing triangle		Black L-shape
Yellow hexagon		Orange circle

Matrix 5

	?	

↓

Options for Matrix 5

1

Yellow hexagon		Black left-pointing triangle
Blue circle		
		Black L-shape

2

Yellow hexagon		Black left-pointing triangle
Blue circle		Black L-shape

3

Yellow hexagon		Black left-pointing triangle
	Blue circle	Black L-shape

Matrix 6

	?	

↓

Options for Matrix 6

Green circle		Yellow hexagon
	Black L-shape	Black upward-pointing triangle

Green circle		Yellow hexagon
Black L-shape		Black upward-pointing triangle

Green circle		Yellow hexagon
Black L-shape		
		Black upward-pointing triangle

Subtest 2 — Mathematical Equations

Each question is a system of equations using letters as unknowns. Find the whole number for every letter so that all equations are satisfied at the same time.

How it works

- Every letter is an integer between 1 and 20.
- There is always exactly one solution that meets all the requirements.
- Solve by substitution and elimination — express letters in terms of one unknown, then back-substitute.
- Score the point only if every letter in the system is correct.

In the real exam: 20 systems in 25 minutes. Be quick and accurate. No notes, no calculator.

Mathematical Equations

Solve the system. Every letter is a whole number from 1 to 20; there is exactly one solution.

$$A + 6 = 13$$

$$B - 4 = A$$

$$A = ?$$

$$B = ?$$

Mathematical Equations

Solve the system. Every letter is a whole number from 1 to 20; there is exactly one solution.

$$2 \times A = 16$$

$$B - A = 5$$

$$A = ?$$

$$B = ?$$

Mathematical Equations

Solve the system. Every letter is a whole number from 1 to 20; there is exactly one solution.

$$A \div 3 = 4$$

$$B = A + 2$$

$$A = ?$$

$$B = ?$$

Mathematical Equations

Solve the system. Every letter is a whole number from 1 to 20; there is exactly one solution.

$$3 \times C = A$$

$$A + C = 12$$

$$A + B = 15$$

$$A = ?$$

$$B = ?$$

$$C = ?$$

Mathematical Equations

Solve the system. Every letter is a whole number from 1 to 20; there is exactly one solution.

$$2 \times A = B$$

$$B + C = 17$$

$$A + C = 11$$

$$A = ?$$

$$B = ?$$

$$C = ?$$

Mathematical Equations

Solve the system. Every letter is a whole number from 1 to 20; there is exactly one solution.

$$A - B + C - D = 7$$

$$4 \times B = C$$

$$3 \times B = A$$

$$8 + B = D$$

$$A = ?$$

$$B = ?$$

$$C = ?$$

$$D = ?$$

Mathematical Equations

Solve the system. Every letter is a whole number from 1 to 20; there is exactly one solution.

$$C + D - A = 2$$

$$5 \times C = D$$

$$12 - C = A$$

$$4 \times C - 2 = B$$

$$A = ?$$

$$B = ?$$

$$C = ?$$

$$D = ?$$

Subtest 3 — Latin Squares

You see a 5×5 grid. Each of the letters A–E appears exactly once in every row and every column. Decide which letter belongs in the highlighted “?” cell.

How it works

- Only the letters shown as response options (A–E) may appear in the grid.
- Use elimination: a cell must take the one letter still missing from its row and column.
- Sometimes you must fill one or more other cells in your mind first before the “?” cell is forced.
- Choose one letter from the response row.

In the real exam: 20 grids in 25 minutes. Be quick and accurate; guess if unsure. No notes.

Latin Squares

Which letter belongs in the highlighted “?” cell? Choose from the response options.

	α	β	γ	δ	ϵ
1	B		E	A	D
2	E		C	?	
3		E		B	A
4	D		A		E
5	A	D	B		C

A	B	C	D	E
---	---	---	---	---

Latin Squares

Which letter belongs in the highlighted “?” cell? Choose from the response options.

	α	β	γ	δ	ϵ
1	?	B		A	E
2				D	
3	E	A	D		B
4	C	D	E	B	
5	B	C	A	E	

A	B	C	D	E
---	---	---	---	---

Latin Squares

Which letter belongs in the highlighted “?” cell? Choose from the response options.

	α	β	γ	δ	ϵ
1					D
2		D		E	
3	E	A	D	B	
4	A			D	
5	D	C	?	A	B

- A
- B
- C
- D
- E

Latin Squares

Which letter belongs in the highlighted “?” cell? Choose from the response options.

	α	β	γ	δ	ϵ
1		D	E	B	
2			?	E	B
3			A		
4			C	D	A
5		C	B	A	D

- A
- B
- C
- D
- E

Latin Squares

Which letter belongs in the highlighted “?” cell? Choose from the response options.

	α	β	γ	δ	ϵ
1	D	C	?		E
2			E		
3		A		D	
4			C		
5	C	B	D	E	

- A
- B
- C
- D
- E

Latin Squares

Which letter belongs in the highlighted “?” cell? Choose from the response options.

	α	β	γ	δ	ϵ
1		D	?		C
2		A			
3	C	B		A	
4	A	E		B	
5	E	C			

A	B	C	D	E
---	---	---	---	---

Answer key

Score one point per question. For Figure Sequences both matrices must be right; for Mathematical Equations every letter must be right.

Q1. M5 = Option 2, M6 = Option 3

Q2. M5 = Option 2, M6 = Option 3

Q3. M5 = Option 1, M6 = Option 1

Q4. M5 = Option 2, M6 = Option 1

Q5. M5 = Option 3, M6 = Option 1

Q6. M5 = Option 2, M6 = Option 2

Q7. M5 = Option 2, M6 = Option 2

Q8. A = 7, B = 11

Q9. A = 8, B = 13

Q10. A = 12, B = 14

Q11. A = 9, B = 6, C = 3

Q12. A = 6, B = 12, C = 5

Q13. A = 9, B = 3, C = 12, D = 11

Q14. A = 10, B = 6, C = 2, D = 10

Q15. D

Q16. D

Q17. E

Q18. D

Q19. A

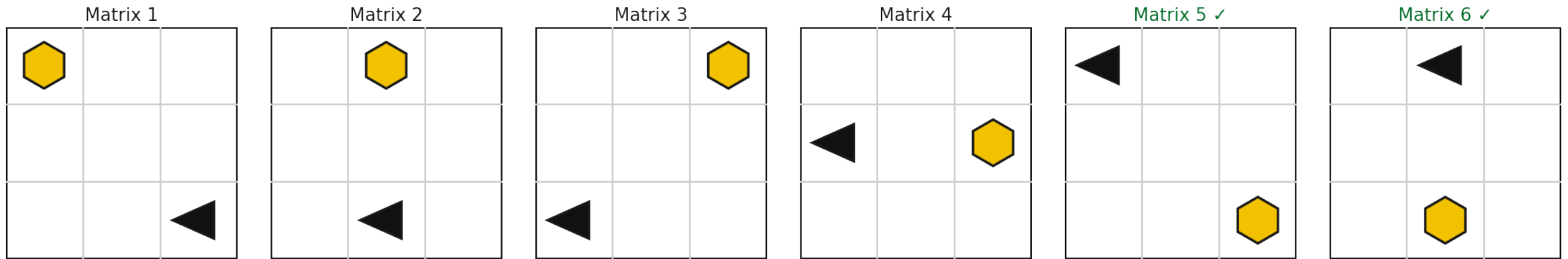
Q20. A

Worked Solutions

Full step-by-step working for all 20 questions. Difficulty is shown here for review.

Figure Sequences

Difficulty: LOW



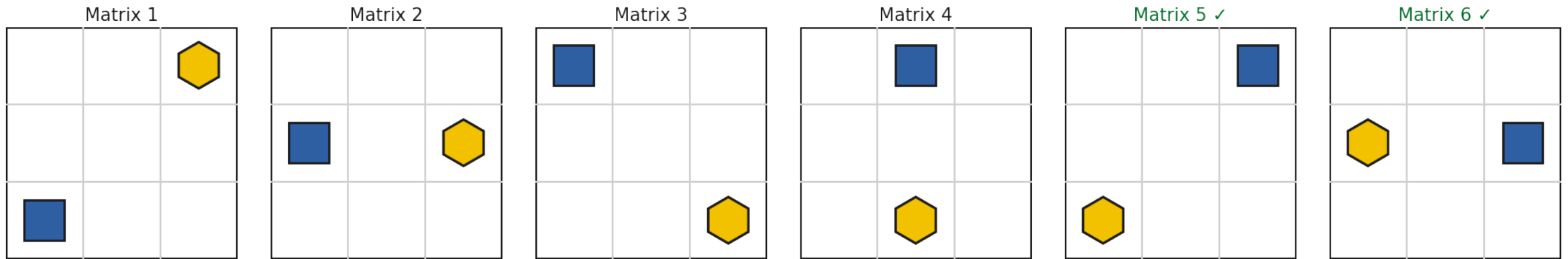
Matrix 5: Option 2 • **Matrix 6: Option 3**

The rules

- Yellow hexagon: moves one field clockwise around the border each matrix.
- Black arrow: moves one field clockwise each matrix; its colour and direction never change.

Figure Sequences

Difficulty: LOW



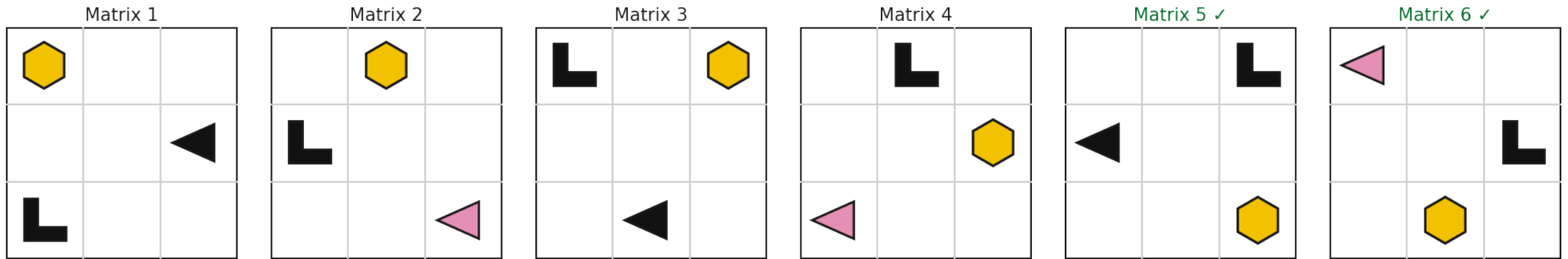
Matrix 5: Option 2 • Matrix 6: Option 3

The rules

- Yellow hexagon: moves one field clockwise around the border each matrix.
- Blue square: moves one field clockwise each matrix; nothing else changes.

Figure Sequences

Difficulty: MEDIUM



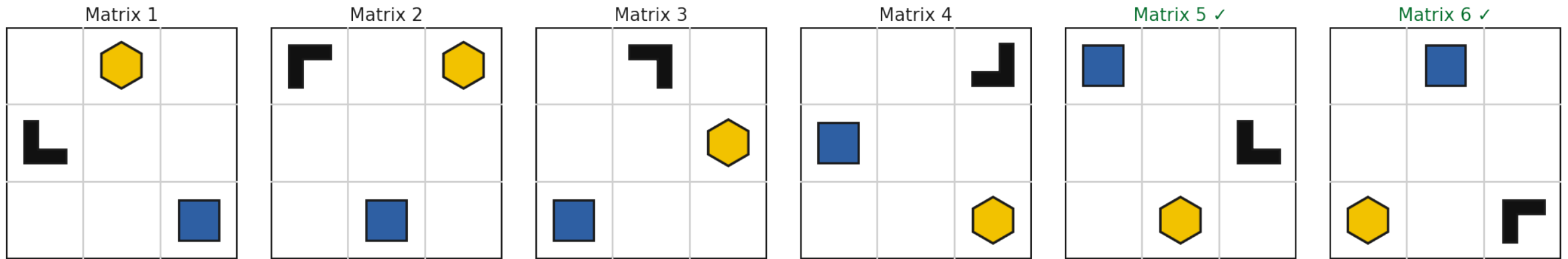
Matrix 5: Option 1 • **Matrix 6: Option 1**

The rules

- Yellow hexagon: moves one field clockwise each matrix.
- Arrow: moves clockwise each matrix and alternates colour black -> pink -> black.
- Bracket: moves one field clockwise each matrix; orientation fixed.

Figure Sequences

Difficulty: MEDIUM



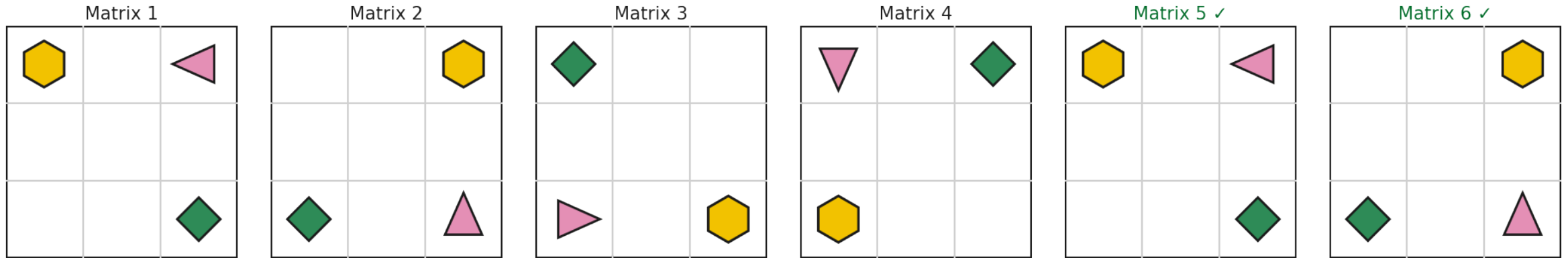
Matrix 5: Option 2 • Matrix 6: Option 1

The rules

- Yellow hexagon: moves one field clockwise each matrix.
- Blue square: moves one field clockwise each matrix.
- Bracket: moves clockwise each matrix and rotates 90 degrees clockwise about its own axis each step.

Figure Sequences

Difficulty: MEDIUM



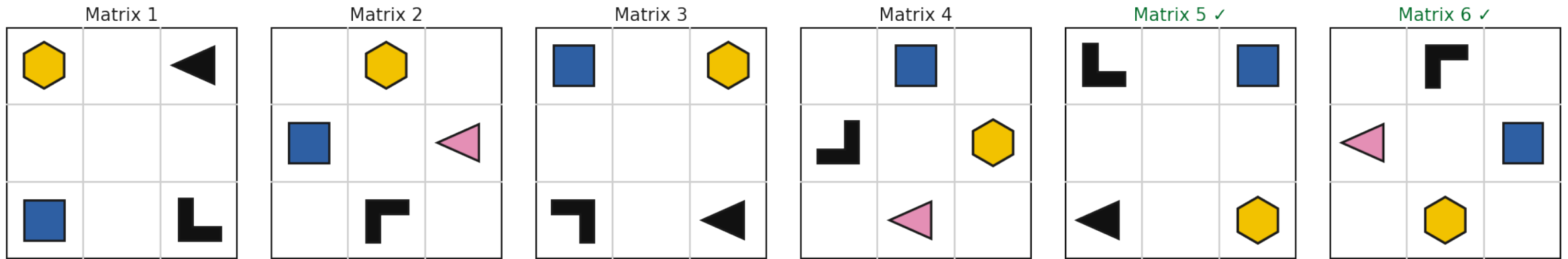
Matrix 5: Option 3 • Matrix 6: Option 1

The rules

- Yellow hexagon: moves two fields clockwise each matrix.
- Pink arrow: moves two fields clockwise each matrix and rotates 90 degrees clockwise each step.
- Green diamond: moves two fields clockwise each matrix.

Figure Sequences

Difficulty: HIGH



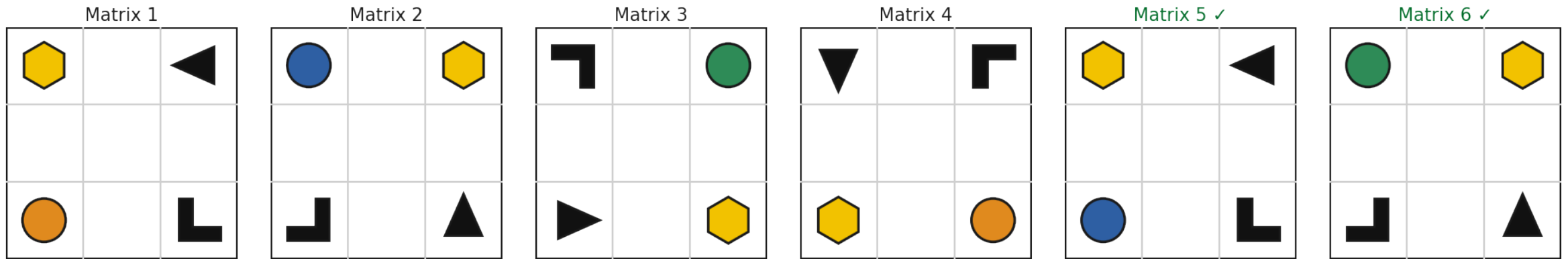
Matrix 5: Option 2 • Matrix 6: Option 2

The rules

- Yellow hexagon: moves one field clockwise each matrix.
- Arrow: moves clockwise and alternates colour black <-> pink each matrix.
- Bracket: moves clockwise and rotates 90 degrees clockwise each matrix.
- Blue square: moves one field clockwise each matrix.

Figure Sequences

Difficulty: HIGH



Matrix 5: Option 2 • Matrix 6: Option 2

The rules

- Yellow hexagon: moves two fields clockwise each matrix.
- Black arrow: moves two fields clockwise and rotates 90 degrees clockwise each matrix.
- Bracket: moves two fields clockwise and rotates 90 degrees counter-clockwise each matrix.
- Circle: moves clockwise and cycles colour orange -> blue -> green each matrix.

Difficulty: LOW

$$A + 6 = 13$$

$$B - 4 = A$$

Answer: $A = 7$, $B = 11$

Step by step

- Equation 1 has only A: $A = 13 - 6 = 7$.
- Substitute into equation 2: $B - 4 = 7 \rightarrow B = 11$.

Mathematical Equations

Difficulty: LOW

$$2 \times A = 16$$

$$B - A = 5$$

Answer: $A = 8$, $B = 13$

Step by step

- Equation 1: $2 \times A = 16 \rightarrow A = 8$.
- Equation 2: $B - 8 = 5 \rightarrow B = 13$.

Difficulty: LOW

$$A \div 3 = 4$$

$$B = A + 2$$

Answer: $A = 12$, $B = 14$

Step by step

- Equation 1: $A \div 3 = 4 \rightarrow A = 12$.
- Equation 2: $B = 12 + 2 = 14$.

Mathematical Equations

Difficulty: MEDIUM

$$3 \times C = A$$

$$A + C = 12$$

$$A + B = 15$$

Answer: $A = 9$, $B = 6$, $C = 3$

Step by step

- $A = 3C$. Into equation 2: $3C + C = 12 \rightarrow 4C = 12 \rightarrow C = 3$, so $A = 9$.
- Equation 3: $9 + B = 15 \rightarrow B = 6$.

Mathematical Equations

Difficulty: MEDIUM

$$2 \times A = B$$

$$B + C = 17$$

$$A + C = 11$$

Answer: $A = 6$, $B = 12$, $C = 5$

Step by step

- Subtract equation 3 from equation 2: $B - A = 6$. With $B = 2A$: $2A - A = 6 \rightarrow A = 6$.
- Then $B = 12$, and from equation 3: $C = 11 - 6 = 5$.

Mathematical Equations

Difficulty: HIGH

$$A - B + C - D = 7$$

$$4 \times B = C$$

$$3 \times B = A$$

$$8 + B = D$$

Answer: $A = 9$, $B = 3$, $C = 12$, $D = 11$

Step by step

- Equations 2-4 give C, A and D in terms of B: $C = 4B$, $A = 3B$, $D = 8 + B$.
- Substitute into equation 1: $3B - B + 4B - (8 + B) = 7 \rightarrow 5B - 8 = 7 \rightarrow 5B = 15 \rightarrow B = 3$.
- Then $A = 9$, $C = 12$, $D = 11$.

Mathematical Equations

Difficulty: HIGH

$$C + D - A = 2$$

$$5 \times C = D$$

$$12 - C = A$$

$$4 \times C - 2 = B$$

Answer: $A = 10$, $B = 6$, $C = 2$, $D = 10$

Step by step

- Equations 2-4 give D, A and B in terms of C: $D = 5C$, $A = 12 - C$, $B = 4C - 2$.
- Substitute into equation 1: $C + 5C - (12 - C) = 2 \rightarrow 7C - 12 = 2 \rightarrow 7C = 14 \rightarrow C = 2$.
- Then $D = 10$, $A = 10$, $B = 6$.

Latin Squares

Difficulty: LOW

	α	β	γ	δ	ϵ
1	B	C	E	A	D
2	E	A	C	D	B
3	C	E	D	B	A
4	D	B	A	C	E
5	A	D	B	E	C

Solution = D

Solution path

- Fill $\beta_1 = C$: only candidate left in its row and column.
- Target $\delta_2 = D$: only candidate left in its row and column.

Latin Squares

Difficulty: LOW

	α	β	γ	δ	ϵ
1	D	B	C	A	E
2	A	E	B	D	C
3	E	A	D	C	B
4	C	D	E	B	A
5	B	C	A	E	D

Solution = D

Solution path

- Target $\alpha 1 = D$: only candidate left in its row and column.

Latin Squares

Difficulty: MEDIUM

	α	β	γ	δ	ϵ
1	B	E	A	C	D
2	C	D	B	E	A
3	E	A	D	B	C
4	A	B	C	D	E
5	D	C	E	A	B

Solution = E

Solution path

- Fill $\delta 1 = C$: only candidate left in its row and column.
- Fill $\epsilon 3 = C$: only candidate left in its row and column.
- Fill $\epsilon 4 = E$: only candidate left in its row and column.
- Target $\gamma 5 = E$: only candidate left in its row and column.

Latin Squares

Difficulty: MEDIUM

	α	β	γ	δ	ϵ
1	A	D	E	B	C
2	C	A	D	E	B
3	D	B	A	C	E
4	B	E	C	D	A
5	E	C	B	A	D

Solution = D

Solution path

- Fill $\epsilon_1 = C$: only candidate left in its row and column.
- Fill $\beta_2 = A$: only candidate left in its row and column.
- Target $\gamma_2 = D$: only candidate left in its row and column.

Latin Squares

Difficulty: HIGH

	α	β	γ	δ	ϵ
1	D	C	A	B	E
2	A	D	E	C	B
3	E	A	B	D	C
4	B	E	C	A	D
5	C	B	D	E	A

Solution = A

Solution path

- Fill $\beta_2 = D$: only candidate left in its row and column.
- Fill $\gamma_3 = B$: only candidate left in its row and column.
- Fill $\epsilon_3 = C$: only candidate left in its row and column.
- Fill $\beta_4 = E$: only candidate left in its row and column.
- Fill $\epsilon_5 = A$: only candidate left in its row and column.
- Target $\gamma_1 = A$: only candidate left in its row and column.

Latin Squares

Difficulty: HIGH

	α	β	γ	δ	ϵ
1	B	D	A	E	C
2	D	A	E	C	B
3	C	B	D	A	E
4	A	E	C	B	D
5	E	C	B	D	A

Solution = A

Solution path

- Fill $\alpha 1 = B$: only candidate left in its row and column.
- Fill $\delta 1 = E$: only candidate left in its row and column.
- Fill $\alpha 2 = D$: only candidate left in its row and column.
- Fill $\delta 2 = C$: only candidate left in its row and column.
- Fill $\epsilon 4 = D$: only candidate left in its row and column.
- Fill $\delta 5 = D$: only candidate left in its row and column.
- Target $\gamma 1 = A$: only candidate left in its row and column.